

Forecasting Austrian Macroeconomic Variables: A Comparative Analysis of ARIMA, VAR, and BVAR Models

1. Introduction

Accurate forecasting of macroeconomic variables is essential for policymakers, businesses, and economists to make informed decisions. This study compares the forecasting performance of three different modeling approaches—ARIMA (AutoRegressive Integrated Moving Average), VAR (Vector Autoregression), and BVAR (Bayesian Vector Autoregression)—for key Austrian macroeconomic indicators.

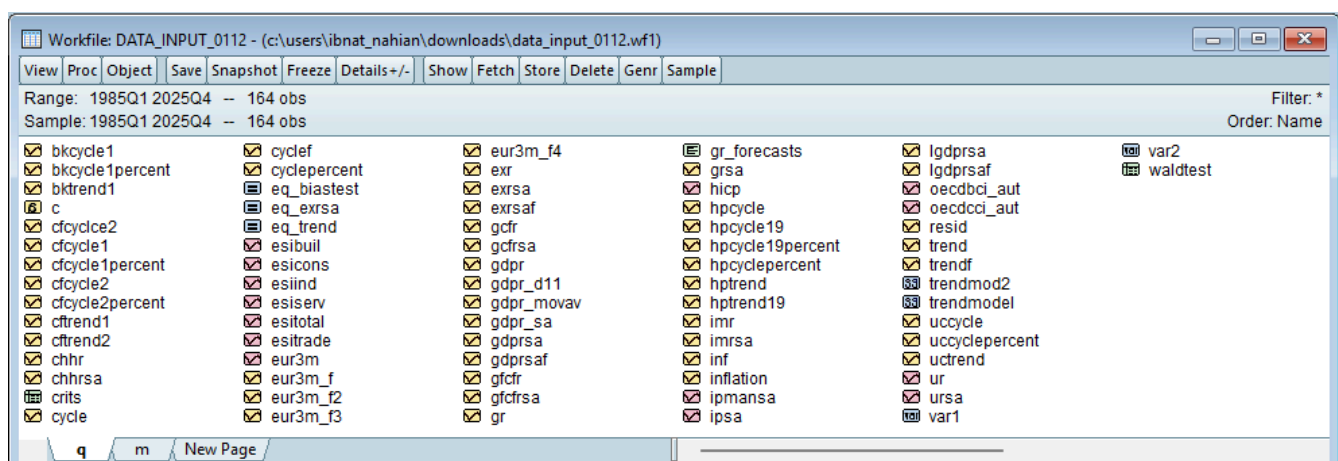
The four variables under investigation are:

- **Unemployment Rate (ursa):** The percentage of the labor force that is unemployed
- **Inflation Rate (inf):** The rate of change in consumer prices
 $genr\ inf = @pcy(hicp)$
- **Real GDP Growth Rate (gdpy):** The percentage change in real economic output year on year
 $genr\ gdpy = @pcy(gdprsa)$
- **Interest Rate (eur3m):** The policy or market interest rate

A key objective of this analysis is to examine whether treating the interest rate as an endogenous variable (determined within the system) versus an exogenous variable (determined outside the system) affects forecast accuracy. This distinction is economically important because interest rates can be viewed both as policy instruments set by central banks (exogenous) and as variables that respond to economic conditions (endogenous).

This study uses Eviews software to find the output. The data spans from 1985Q1 to 2025Q4, providing 164 quarterly observations. We perform a pseudo out-of-sample forecast evaluation using a recursive estimation window. The first forecast origin is 1997Q4. For each forecast origin t_0 we estimate the model using data up to t_0 and produce h -step ahead forecasts for horizons $h = 1, 2, 4, 8$. We then advance the origin by one quarter and repeat until the final origin (2025Q2). RMSE and MAE are computed by averaging squared/absolute errors across all origins for each horizon and then averaging across horizons when we report an overall value.

The following is the workfile used-



2. Data Description and Preliminary Analysis

2.1 Data Sources and Variables

All data were obtained from official statistical sources for Austria, likely Eurostat. The quarterly frequency is standard for macroeconomic analysis, providing sufficient observations while capturing business cycle dynamics. The Eviews 14 and Eviews 12 software was used primarily to find all the results. Occasionally, ChatGPT 5.2 was used to assist with the analysis.

2.2 Unit Root Tests and Stationarity

Before estimating time-series models, it is essential to examine whether the variables are stationary. A stationary time series is characterized by a constant mean, constant variance, and autocovariances that depend only on the lag length and not on time itself. The use of non-stationary variables may lead to spurious regressions and unreliable forecasting results.

To assess stationarity, the **Augmented Dickey–Fuller (ADF)** test and the **Kwiatkowski–Phillips–Schmidt–Shin (KPSS)** test were conducted for each variable. The null hypothesis of the ADF test is that the series contains a unit root (non-stationary), whereas the null hypothesis of the KPSS test is that the series is stationary. Using both tests jointly allows for a more robust assessment of the integration properties of the variables.

Key Findings from Unit Root test:

Unemployment Rate (ursa): ADF suggests stationarity, KPSS also does not reject stationarity (after first - differencing)

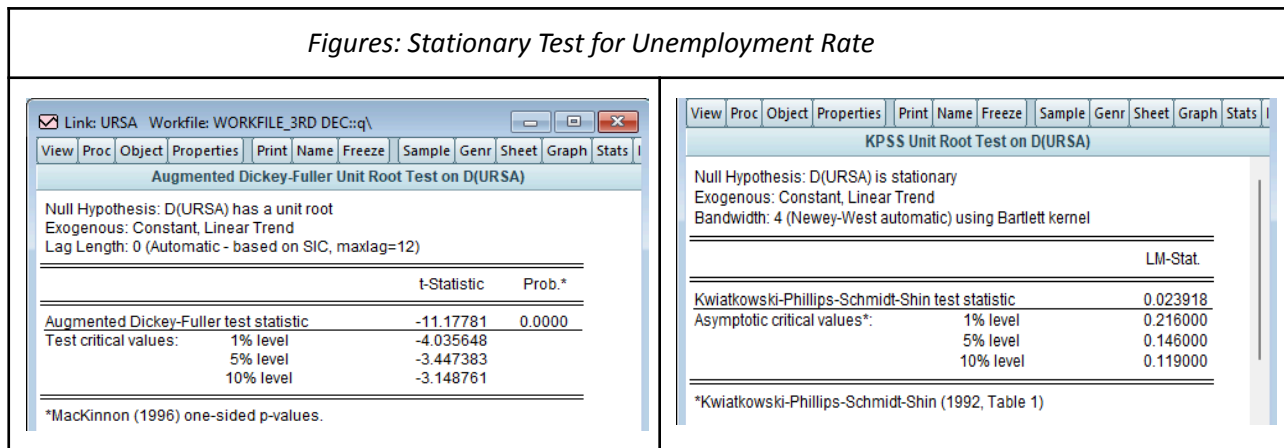
Inflation Rate (inf): ADF suggests stationarity, KPSS also does not reject stationarity at levels

GDP Growth Rate (gdp): ADF suggests stationarity, KPSS also does not reject stationarity at levels

Interest Rate (eur3m): ADF suggests stationarity, KPSS also does not reject stationarity (after first-differencing)

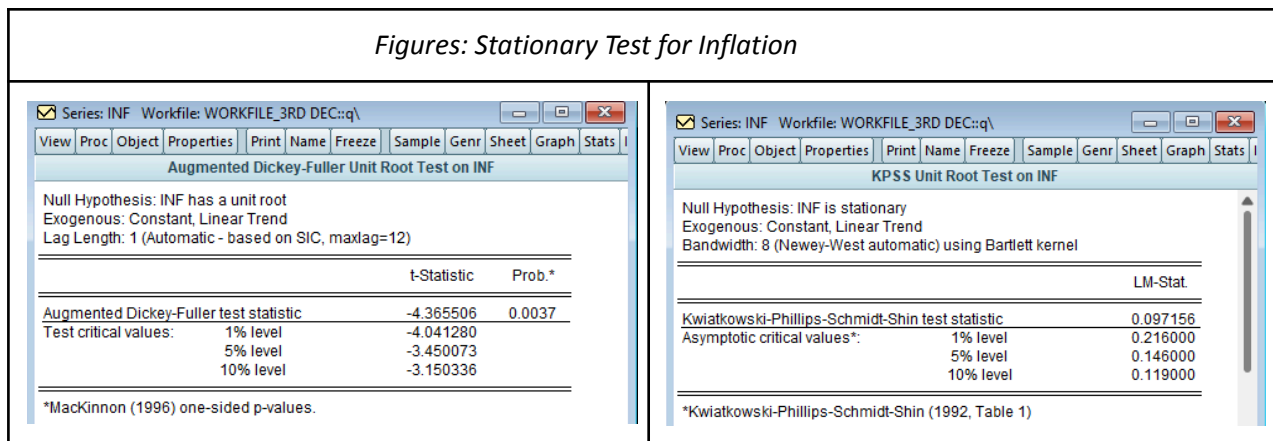
Overall, the results suggest that the inflation rate and GDP growth rate can be treated as **I(0)** variables, while the unemployment rate and interest rate are **I(1)** processes. These findings are consistent with standard macroeconomic theory and guide the specification of the ARIMA, VAR, and BVAR models in the subsequent analysis. The results are below:

Figures: Stationary Test for Unemployment Rate



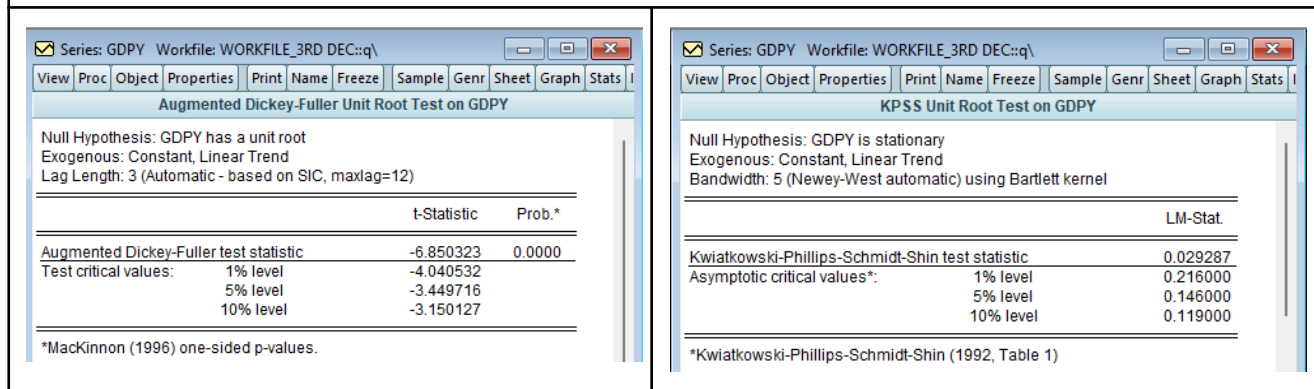
Interpretation: The Augmented Dickey–Fuller test is conducted under the null hypothesis that the series contains a unit root. For the first-differenced unemployment rate, the ADF test statistic is -11.1778 , which is more negative than the critical values at the 1%, 5%, and 10% significance levels. The associated p-value is 0.0000, indicating strong statistical significance. Accordingly, the null hypothesis of a unit root is rejected at the 1%, 5%, and 10% levels of significance. The KPSS test, which has stationarity as the null hypothesis, yields a test statistic of 0.0239, which is well below the critical values at the 1%, 5%, and 10% significance levels. Therefore, the null hypothesis of stationarity cannot be rejected at any conventional significance level. The combined evidence from the ADF and KPSS tests indicates that the unemployment rate is non-stationary in levels but stationary after first differencing, implying that it is an $I(1)$ process.

Figures: Stationary Test for Inflation



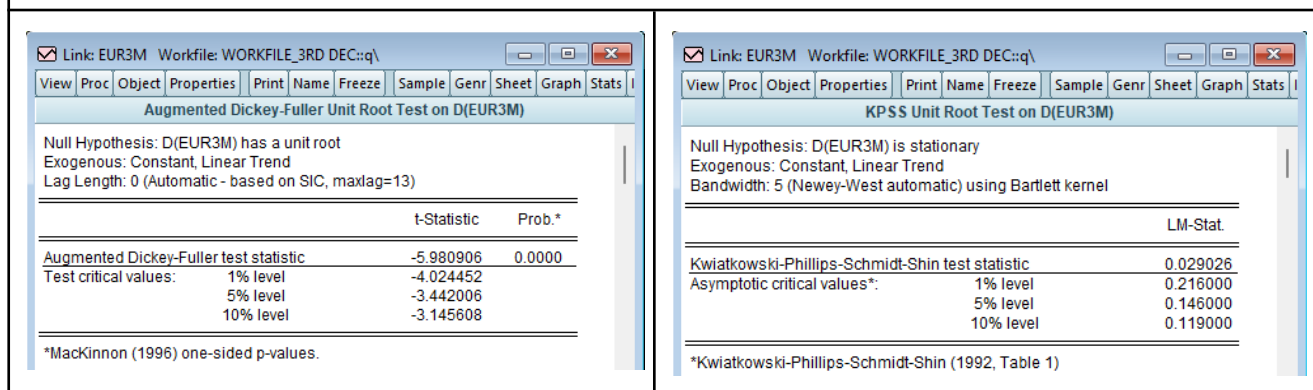
Interpretation: The stationarity properties of the inflation rate were examined using both the Augmented Dickey–Fuller (ADF) test and the KPSS test, each including a constant and a linear time trend. The ADF test rejects the null hypothesis of a unit root, with a test statistic of -4.3655 , which is more negative than the critical values at the 1%, 5%, and 10% significance levels, and a corresponding p-value of 0.0037. This indicates that the inflation rate does not exhibit unit root behavior. Consistent with this result, the KPSS test yields a test statistic of 0.0972, which lies below the critical values at the 1%, 5%, and 10% levels, implying that the null hypothesis of stationarity cannot be rejected. Taken together, the results from both tests provide strong evidence that the inflation rate is stationary in levels, indicating that inflation can be treated as an $I(0)$ process in the subsequent econometric analysis.

Figures: Stationary Test for year on year GDP



Interpretation: The ADF test strongly rejects the null hypothesis of a unit root, with a test statistic of -6.8503 , which is more negative than the critical values at the 1%, 5%, and 10% significance levels, and a corresponding p-value of 0.0000. This indicates that real GDP growth does not exhibit unit root behavior. Consistent with this finding, the KPSS test yields a test statistic of 0.0293, which lies well below the critical values at all conventional significance levels, implying that the null hypothesis of stationarity cannot be rejected. Taken together, the results from both tests provide robust evidence that the real GDP growth rate is stationary in levels, indicating that GDP growth can be treated as an $I(0)$ process in the subsequent econometric and forecasting analysis.

Figures: Stationary Test for Interest Rate



Interpretation: The stationarity properties of the short-term interest rate were examined using the Augmented Dickey-Fuller (ADF) test and the KPSS test, both specified with a constant and a linear time trend. The ADF test strongly rejects the null hypothesis of a unit root for the first-differenced interest rate, with a test statistic of -5.9809 , which is more negative than the critical values at the 1%, 5%, and 10% significance levels, and an associated p-value of 0.0000. This indicates that the interest rate becomes stationary after first differencing. Consistent with this result, the KPSS test yields a statistic of 0.0290, which lies well below the critical values at all conventional significance levels, implying that the null hypothesis of stationarity cannot be rejected for the differenced series. Taken together, these findings suggest that the interest rate is non-stationary in levels but stationary in first differences, indicating that EUR3M can be characterized as an $I(1)$ process.

2.3 Visual Inspection

Visual inspection of the time series reveals distinct dynamic patterns across the macroeconomic variables.

Figure: Unemployment Rate

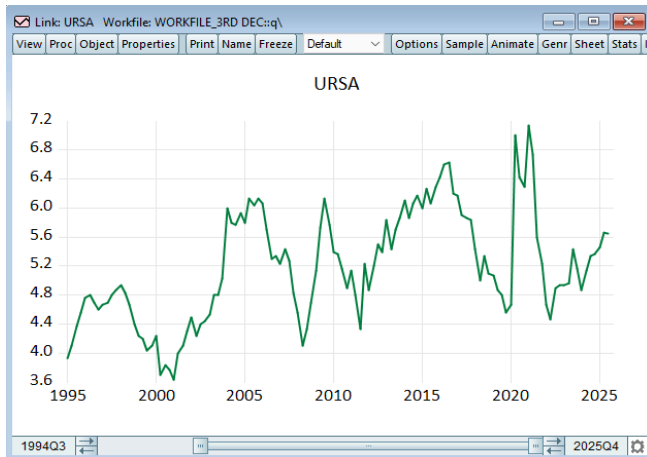


Figure: Inflation Rate

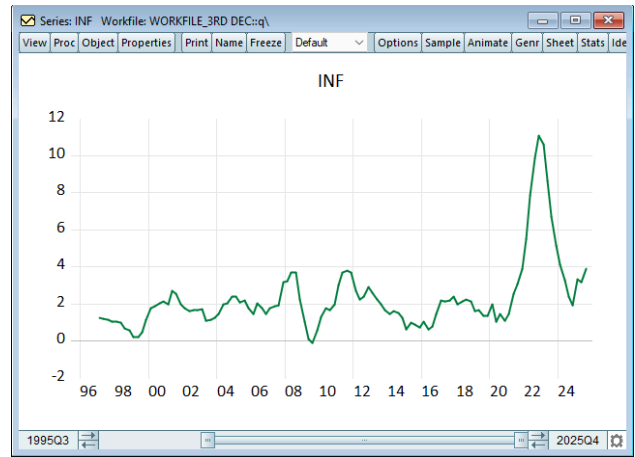


Figure: GDP Growth Rate

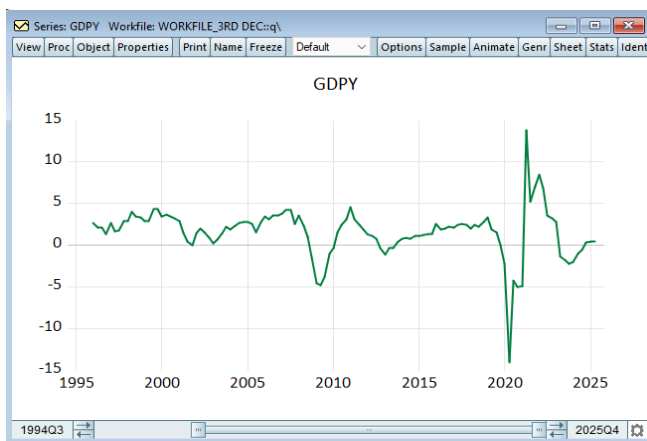
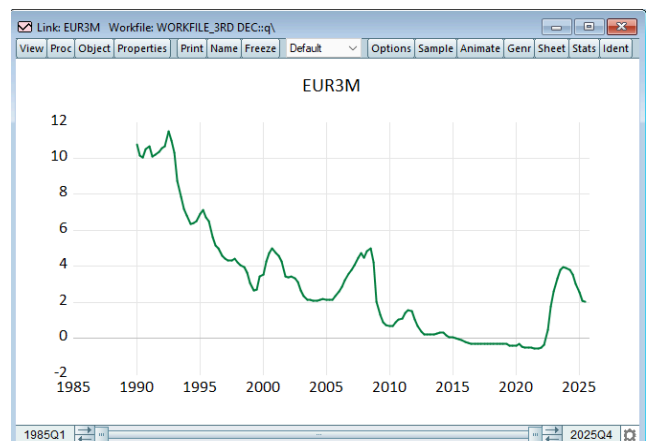


Figure: Interest Rate



The unemployment rate displays pronounced cyclical movements and persistence, with noticeable increases during major economic downturns, suggesting non-stationarity in levels. Inflation fluctuates around a relatively stable mean for most of the sample period, despite episodes of heightened volatility, particularly in recent years, indicating mean-reverting behavior consistent with stationarity. Real GDP growth exhibits strong cyclical fluctuations with sharp contractions during economic crises followed by recoveries, but no evident long-term trend, supporting stationarity in levels. In contrast, the short-term interest rate shows a clear downward trend over most of the sample, remaining near the zero lower bound for an extended period before rising sharply in recent years, which is indicative of non-stationary behavior in levels. Overall, the visual evidence is consistent with the formal unit root test results and supports the chosen data transformations for the subsequent econometric analysis.

3. Methodology

3.1 ARIMA Models

Univariate ARIMA models were estimated for each macroeconomic variable using maximum likelihood estimation, with model selection based on the Akaike and Schwarz information criteria. The selected specifications are stable, as indicated by inverted AR and MA roots lying within the unit circle. An ARIMA(p,d,q) model has three components:

AR(p) - Autoregressive part: The variable depends on its own past p values

I(d) - Integrated part: The variable is differenced d times to make it stationary

MA(q) - Moving Average part: The variable depends on past q forecast errors

Figures: Arima Model Forecasting Equations

Figures: Unemployment Rate

	A	B	C
155	(3,2)(0,0)	0.882728	
156	(3,2)(0,0)	0.882728	
157	(3,2)(0,4)	0.794770	
158	(3,2)(0,0)	0.882728	
159	(3,2)(0,0)	0.882728	
160	(3,2)(0,4)	0.794770	
161	(3,2)(4,0)	0.884684	
162	(3,2)(4,0)	0.884684	
163	(3,2)(4,4)	0.768611	
164	(3,3)(0,0)	0.860365	
165	(3,3)(0,0)	0.860365	
166	(3,3)(0,4)	0.840371	
167	(3,3)(0,0)	0.860365	
168	(3,3)(0,0)	0.860365	
169	(3,3)(0,4)	0.840371	
170	(3,3)(4,0)	0.854029	
171	(3,3)(4,0)	0.854029	
172	(3,3)(4,4)	0.783314	
173	(3,4)(0,0)	0.796929	
174	(3,4)(0,0)	0.796929	
175	(3,4)(0,4)	0.810901	
176	(3,4)(0,0)	0.796929	
177	(3,4)(0,0)	0.796929	
178	(3,4)(0,4)	0.810901	
179	(3,4)(4,0)	NA	
180	(3,4)(4,0)	NA	
181	(3,4)(4,4)	0.797065	
182	(4,0)(0,0)	0.872253	

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.011183	0.005454	2.050471	0.0426
AR(1)	0.425199	0.171704	2.476348	0.0148
AR(2)	-0.921501	0.086955	-10.59745	0.0000
AR(3)	0.285680	0.144308	1.979659	0.0502
SAR(4)	0.332730	0.166283	2.000994	0.0478
MA(1)	-0.446952	0.148432	-3.011144	0.0032
MA(2)	0.740767	0.129082	5.738739	0.0000
SMA(4)	-0.999998	212.5690	-0.004704	0.9963
SIGMASQ	0.100851	10.91329	0.009241	0.9926

R-squared	0.270933	Mean dependent var	0.014071
Adjusted R-squared	0.219317	S.D. dependent var	0.373460
S.E. of regression	0.329975	Akaike info criterion	0.768611
Sum squared resid	12.30383	Schwarz criterion	0.975465
Log likelihood	-37.88525	Hannan-Quinn criter.	0.852628
F-statistic	5.249070	Durbin-Watson stat	2.011973
Prob(F-statistic)	0.000013		

Inverted AR Roots	.76	.32	.05+.94i	.05-.94i
	.00+.76i	-.00-.76i	-.76	
Inverted MA Roots	1.00	.22-.83i	.22+.83i	-.00+1.00i
	-.00-1.00i	-1.00		

The value 0.768611 is the lowest information criterion value for the unemployment rate series, indicating that the corresponding ARIMA specification provides the best balance between goodness of fit and parsimony and is therefore selected for unemployment forecasting. For the unemployment rate, which is non-stationary in levels, an ARIMA model in first differences was selected. The preferred specification includes multiple autoregressive and moving-average terms, capturing the persistence and cyclical dynamics observed in unemployment. Most coefficients are statistically significant, and the model diagnostics indicate a satisfactory fit, making it suitable for forecasting purposes.

Figures: Inflation Rate

Table: CRITS10 Workfile: ...				
View	Proc	Object	Print	Name
		A	B	C
63		(1,1)(4,0)	1.723121	
64		(1,1)(4,4)	1.649708	
65		(1,2)(0,0)	1.687859	
66		(1,2)(0,0)	1.687859	
67		(1,2)(0,4)	1.530635	
68		(1,2)(0,0)	1.687859	
69		(1,2)(0,0)	1.687859	
70		(1,2)(0,4)	1.530635	
71		(1,2)(4,0)	1.638255	
72		(1,2)(4,0)	1.638255	
73		(1,2)(4,4)	1.529672	
74		(1,3)(0,0)	1.408471	
75		(1,3)(0,0)	1.408471	
76		(1,3)(0,4)	1.406982	
77		(1,3)(0,0)	1.408471	
78		(1,3)(0,0)	1.408471	
79		(1,3)(0,4)	1.406982	
80		(1,3)(4,0)	1.412512	
81		(1,3)(4,0)	1.412512	
82		(1,3)(4,4)	1.385111	
83		(1,4)(0,0)	1.422782	
84		(1,4)(0,0)	1.422782	
85		(1,4)(0,4)	1.422656	

Series: INF Workfile: DATA_INPUT_0112::q\				
View	Proc	Object	Print	Name
Dependent variable: INF Method: ARIMA Maximum Likelihood (BFGS) Date: 12/04/25 Time: 15:57 Sample: 1997Q1 2025Q3 Included observations: 115 Convergence achieved after 236 iterations Coefficient covariance computed using outer product of gradients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.286447	0.608639	3.756656	0.0003
AR(1)	0.674609	0.067515	9.992051	0.0000
SAR(4)	-0.739729	0.153782	-4.810249	0.0000
MA(1)	0.827152	88.75209	0.009320	0.9926
MA(2)	1.020403	195.3117	0.005224	0.9958
MA(3)	0.801694	239.5890	0.003346	0.9973
SMA(4)	0.922804	0.141791	6.508201	0.0000
SIGMASQ	0.183989	4.459189	0.041261	0.9672
R-squared	0.949588	Mean dependent var	2.308770	
Adjusted R-squared	0.946290	S.D. dependent var	1.918780	
S.E. of regression	0.444686	Akaike info criterion	1.385111	
Sum squared resid	21.15876	Schwarz criterion	1.576062	
Log likelihood	-71.64386	Hannan-Quinn criter.	1.462617	
F-statistic	287.9295	Durbin-Watson stat	2.027582	
Prob(F-statistic)	0.000000			
Inverted AR Roots	.67	.66-.66i	.66-.66i	-.66+.66i
Inverted MA Roots	.69+.69i	.69+.69i	-.01-1.00i	-.01+1.00i
	-.69-.69i	-.69-.69i	-.80	

The value 1.385111 is the lowest information criterion value for the inflation series, indicating that the corresponding ARIMA specification provides the best fit among the tested models after accounting for model complexity and is therefore selected for inflation forecasting. The inflation rate, which is stationary in levels, is modeled using an ARIMA specification without differencing. The selected model includes both AR and MA components, reflecting short-run persistence and temporary shocks in inflation dynamics. The estimated coefficients are statistically significant, and the relatively low information criteria values suggest a good in-sample fit.

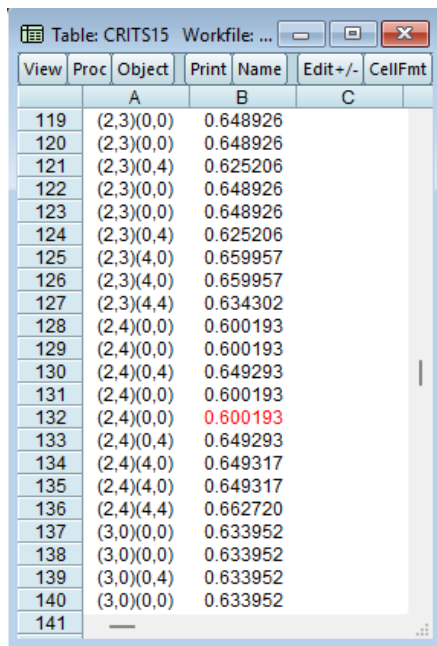
Figures: Year on Year GDP Growth

Table: CRITS03 Workfile: ...				
View	Proc	Object	Print	Name
		A	B	C
154		(3,1)(4,4)	4.101509	
155		(3,2)(0,0)	4.249532	
156		(3,2)(0,0)	4.249532	
157		(3,2)(0,4)	4.047926	
158		(3,2)(0,0)	4.249532	
159		(3,2)(0,0)	4.249532	
160		(3,2)(0,4)	4.047926	
161		(3,2)(4,0)	4.206008	
162		(3,2)(4,0)	4.206008	
163		(3,2)(4,4)	3.993634	
164		(3,3)(0,0)	4.079357	
165		(3,3)(0,0)	4.079357	
166		(3,3)(0,4)	3.999793	
167		(3,3)(0,0)	4.079357	
168		(3,3)(0,0)	4.079357	
169		(3,3)(0,4)	3.999793	
170		(3,3)(4,0)	NA	
171		(3,3)(4,0)	NA	
172		(3,3)(4,4)	4.010583	
173		(3,4)(0,0)	4.100637	
174		(3,4)(0,0)	4.100637	

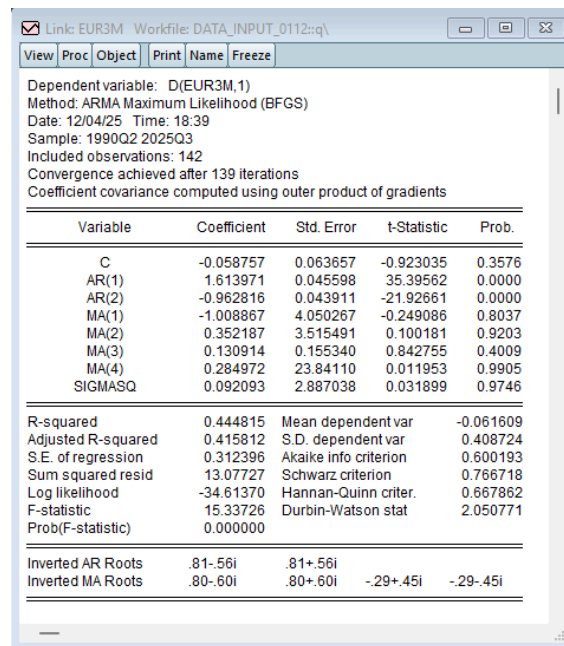
Series: GDPY Workfile: DATA_INPUT_0112 (2)::q\				
View	Proc	Object	Print	Name
Dependent variable: GDPY Method: ARIMA Maximum Likelihood (BFGS) Date: 12/15/25 Time: 14:40 Sample: 1996Q1 2025Q2 Included observations: 118 Convergence achieved after 34 iterations Coefficient covariance computed using outer product of gradients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.601111	0.225908	7.087453	0.0000
AR(1)	0.642197	0.072662	8.838143	0.0000
AR(2)	-0.522928	0.139885	-3.738274	0.0003
AR(3)	0.689055	0.130152	5.294246	0.0000
SAR(4)	0.114162	0.134500	0.848790	0.3979
MA(1)	-0.010593	0.084774	-0.124958	0.9008
MA(2)	0.978106	0.365952	2.672772	0.0087
SMA(4)	-0.999999	569.3596	-0.001756	0.9986
SIGMASQ	2.244462	641.2562	0.003500	0.9972
R-squared	0.735692	Mean dependent var	1.617732	
Adjusted R-squared	0.716293	S.D. dependent var	2.926503	
S.E. of regression	1.558777	Akaike info criterion	3.993634	
Sum squared resid	264.8456	Schwarz criterion	4.204958	
Log likelihood	-226.6244	Hannan-Quinn criter.	4.079438	
F-statistic	37.92459	Durbin-Watson stat	1.985106	
Prob(F-statistic)	0.000000			
Inverted AR Roots	.91	.58	.00-.58i	-.00+.58i
Inverted MA Roots	-.13-.86i	-.13+.86i	-.58	
	1.00	.01+.99i	.01-.99i	-.00+1.00i
	-.00-1.00i	-1.00		

Similarly, the real GDP growth rate, which is also stationary in levels, is modeled using an ARIMA process without differencing. The selected specification captures the strong cyclical behavior of output growth through several autoregressive and moving-average terms. The model successfully accounts for short-term fluctuations and volatility associated with business cycle movements. The value 3.993634 is the lowest information criterion value for the GDP growth series, indicating that the corresponding ARIMA specification offers the best balance between goodness of fit and model parsimony and is therefore selected for forecasting real GDP growth.

Figures: Interest Rate



	A	B	C
119	(2,3)(0,0)	0.648926	
120	(2,3)(0,0)	0.648926	
121	(2,3)(0,4)	0.625206	
122	(2,3)(0,0)	0.648926	
123	(2,3)(0,0)	0.648926	
124	(2,3)(0,4)	0.625206	
125	(2,3)(4,0)	0.659957	
126	(2,3)(4,0)	0.659957	
127	(2,3)(4,4)	0.634302	
128	(2,4)(0,0)	0.600193	
129	(2,4)(0,0)	0.600193	
130	(2,4)(0,4)	0.649293	
131	(2,4)(0,0)	0.600193	
132	(2,4)(0,0)	0.600193	
133	(2,4)(0,4)	0.649293	
134	(2,4)(4,0)	0.649317	
135	(2,4)(4,0)	0.649317	
136	(2,4)(4,4)	0.662720	
137	(3,0)(0,0)	0.633952	
138	(3,0)(0,0)	0.633952	
139	(3,0)(0,4)	0.633952	
140	(3,0)(0,0)	0.633952	
141	—		



Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.058757	0.063657	-0.923035	0.3576
AR(1)	1.613971	0.045598	35.39562	0.0000
AR(2)	-0.962816	0.043911	-21.92661	0.0000
MA(1)	-1.008867	4.050267	-0.249086	0.8037
MA(2)	0.352187	3.515491	0.100181	0.9203
MA(3)	0.130914	0.155340	0.842755	0.4009
MA(4)	0.284972	23.84110	0.011953	0.9905
SIGMASQ	0.092093	2.887038	0.031899	0.9746

R-squared	0.444815	Mean dependent var	-0.061609
Adjusted R-squared	0.415812	S.D. dependent var	0.408724
S.E. of regression	0.312396	Akaike info criterion	0.600193
Sum squared resid	13.07727	Schwarz criterion	0.766718
Log likelihood	-34.61370	Hannan-Quinn criter.	0.667862
F-statistic	15.33726	Durbin-Watson stat	2.050771
Prob(F-statistic)	0.000000		

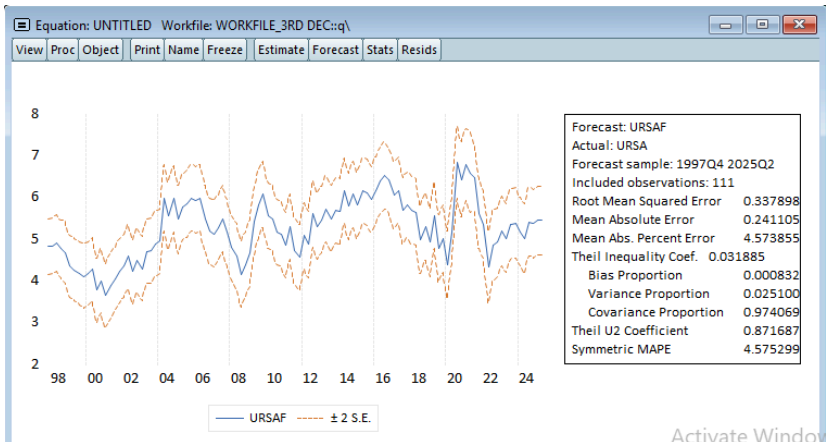
Inverted AR Roots	.81-.56i	.81+.56i		
Inverted MA Roots	.80-.60i	.80+.60i	-.29+.45i	-.29-.45i

Finally, for the short-term interest rate, which is found to be integrated of order one, an ARIMA model in first differences is estimated. The chosen specification reflects the persistence and adjustment dynamics of interest rate changes, particularly during periods of monetary policy shifts. Diagnostic statistics indicate model stability and adequacy for forecasting. The value 0.600193 is the minimum information criterion value for the interest rate series, indicating that the corresponding ARIMA specification provides the best fit among the tested models after accounting for model complexity and is therefore selected for interest rate forecasting.

For Forecasting, samples from 1997q4 to 2025q2 are selected.

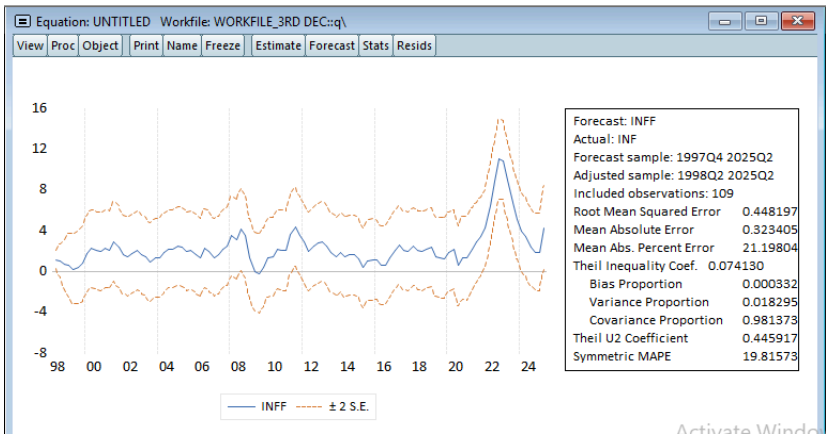
Figures: ARIMA Model Forecasting

Unemployment rate: $d(ursa) c ar(1) ar(2) ar(3) sar(4) ma(1) ma(2) sma(4)$



The ARIMA model for the unemployment rate produces forecasts that broadly track the actual series over the evaluation period, capturing the main cyclical movements while smoothing short-term volatility. The forecast accuracy measures indicate a reasonable performance, with a Root Mean Squared Error (RMSE) of 0.3378 and a Mean Absolute Error (MAE) of 0.2411, suggesting relatively small average forecast errors. The Theil’s inequality coefficient is 0.0318, which is close to zero and indicates good forecasting performance relative to a naïve benchmark. The error decomposition shows that most of the forecast error is attributed to the covariance proportion, implying that errors mainly arise from imperfect tracking of turning points rather than systematic bias or variance mismatch. Overall, the ARIMA model provides a satisfactory univariate benchmark for forecasting unemployment, against which the multivariate VAR and BVAR models can be compared.

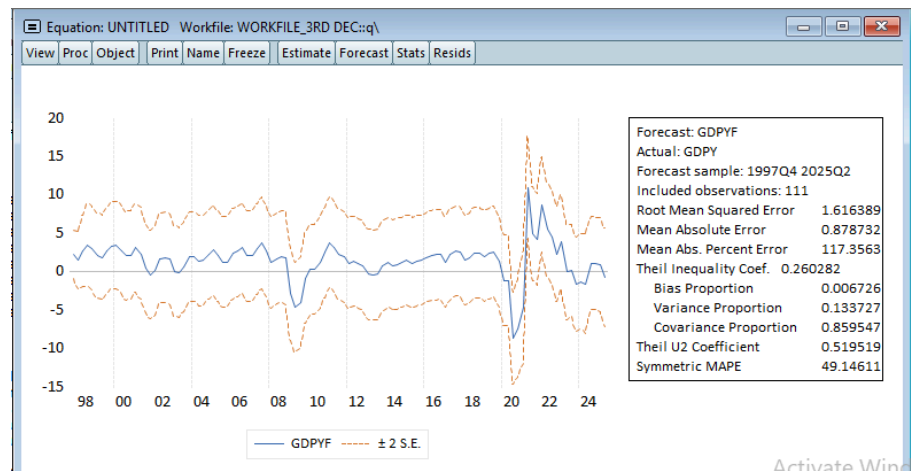
Inflation rate: $inf c ar(1) sar(4) ma(1) ma(2) ma(3) sma(4)$



The ARIMA model for the inflation rate captures the general movements of inflation over the forecast evaluation period but shows weaker performance during episodes of sharp volatility, particularly in the post-pandemic period. The forecast accuracy measures indicate moderate predictive accuracy, with a Root Mean Squared Error (RMSE) of 0.4481 and a Mean Absolute Error (MAE) of 0.3234. The Theil’s inequality coefficient equals 0.0741, which remains relatively low but is higher than that observed for the unemployment rate, indicating comparatively lower forecast precision. The error decomposition reveals that the majority of the forecast error is attributed to the covariance proportion, suggesting that inaccuracies mainly stem from difficulties in timing rapid inflationary movements rather than systematic bias. Overall, while the ARIMA model

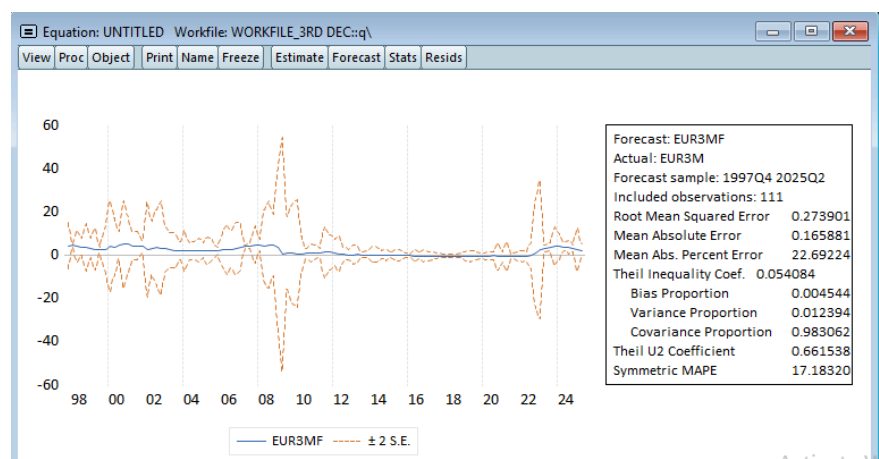
provides a reasonable benchmark forecast for inflation, its performance deteriorates during periods of extreme shocks, highlighting the potential advantages of multivariate forecasting frameworks.

GDP growth: $gdp_y \sim c \ar(1) \ar(2) \ar(3) sar(4) ma(1) ma(2) sma(4)$



The ARIMA model for real GDP growth captures the overall cyclical behavior of output growth but struggles to accurately predict periods of extreme economic disruption. The forecast evaluation statistics indicate relatively weaker performance compared to unemployment and inflation, with a Root Mean Squared Error (RMSE) of 1.6164 and a Mean Absolute Error (MAE) of 0.8787. The Theil's inequality coefficient equals 0.2603, suggesting a notable deviation from actual outcomes, particularly during sharp contractions and rebounds. The error decomposition shows that the majority of forecast errors arise from the covariance proportion, indicating difficulties in correctly timing turning points rather than systematic bias. Overall, while the ARIMA model provides a useful univariate benchmark for GDP growth forecasting, its limited ability to anticipate large shocks highlights the potential benefits of incorporating additional macroeconomic information through multivariate models.

Interest rate: $d(eur3m) \sim c \ar(1) \ar(2) ma(1) ma(2) ma(3) ma(4)$



The ARIMA model for the short-term interest rate provides a reasonable approximation of the overall dynamics of interest rate changes but exhibits limited precision during periods of heightened volatility. The forecast accuracy measures indicate moderate performance, with a Root Mean Squared Error (RMSE) of 0.2739 and a Mean Absolute Error (MAE) of 0.165. The Theil's inequality coefficient is 0.0541, suggesting relatively good forecasting accuracy compared to a naïve benchmark. The error decomposition shows that most of the forecast

error is driven by the covariance proportion, indicating that inaccuracies primarily stem from timing issues rather than systematic bias. Overall, the ARIMA model serves as an adequate univariate benchmark for interest rate forecasting, though its performance may be improved by incorporating additional macroeconomic information in multivariate models.

Advantage: ARIMA models are simple, widely used, and can capture complex patterns in individual series through careful selection of p , d , and q .

Limitation: ARIMA models ignore relationships between variables. For example, they don't capture how changes in interest rates might affect unemployment or GDP growth.

3.2 VAR Models

VAR models are multivariate models that treat all variables as endogenous and allow each variable to depend on past values of itself and all other variables in the system. A VAR(p) model includes p lags of each variable.

For example, in a VAR(2) model with four variables, the unemployment rate at time t depends on:

- Unemployment at $t-1$ and $t-2$
- Inflation at $t-1$ and $t-2$
- GDP growth at $t-1$ and $t-2$
- Interest rate at $t-1$ and $t-2$

Advantage: VAR models capture the dynamic interactions between variables, which is crucial for macroeconomic analysis where variables influence each other.

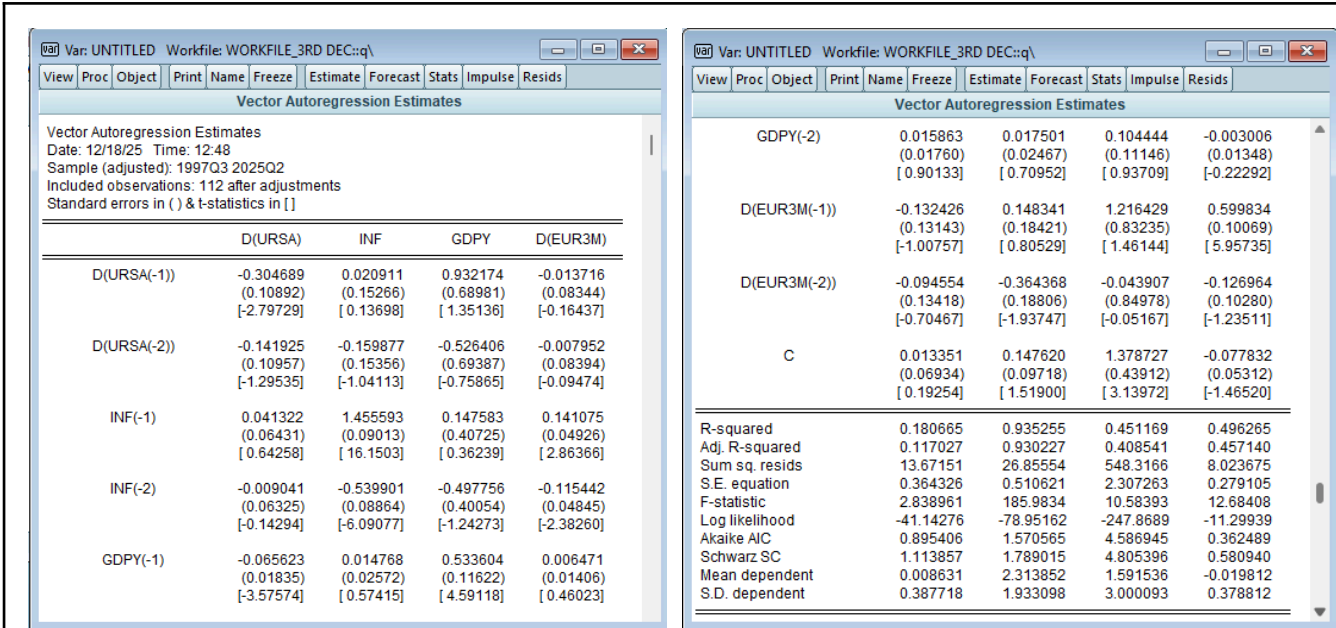
Limitation: VAR models require estimating many parameters (with 4 variables and 2 lags, each equation has 8 coefficients plus a constant). This can lead to imprecise estimates, especially with limited data.

Two VAR Specification:

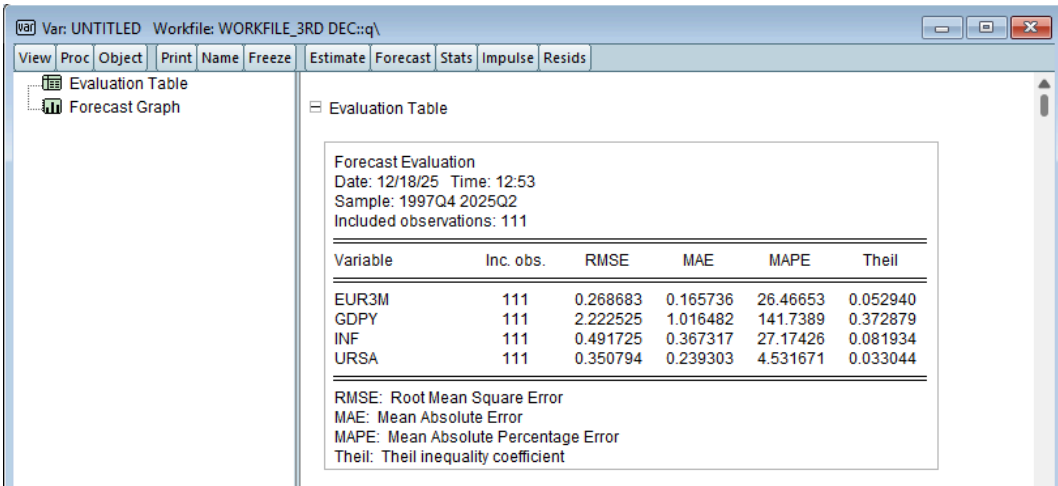
All Endogenous: All four variables (unemployment, inflation, GDP growth, interest rate) are treated as endogenous, meaning they all influence each other.

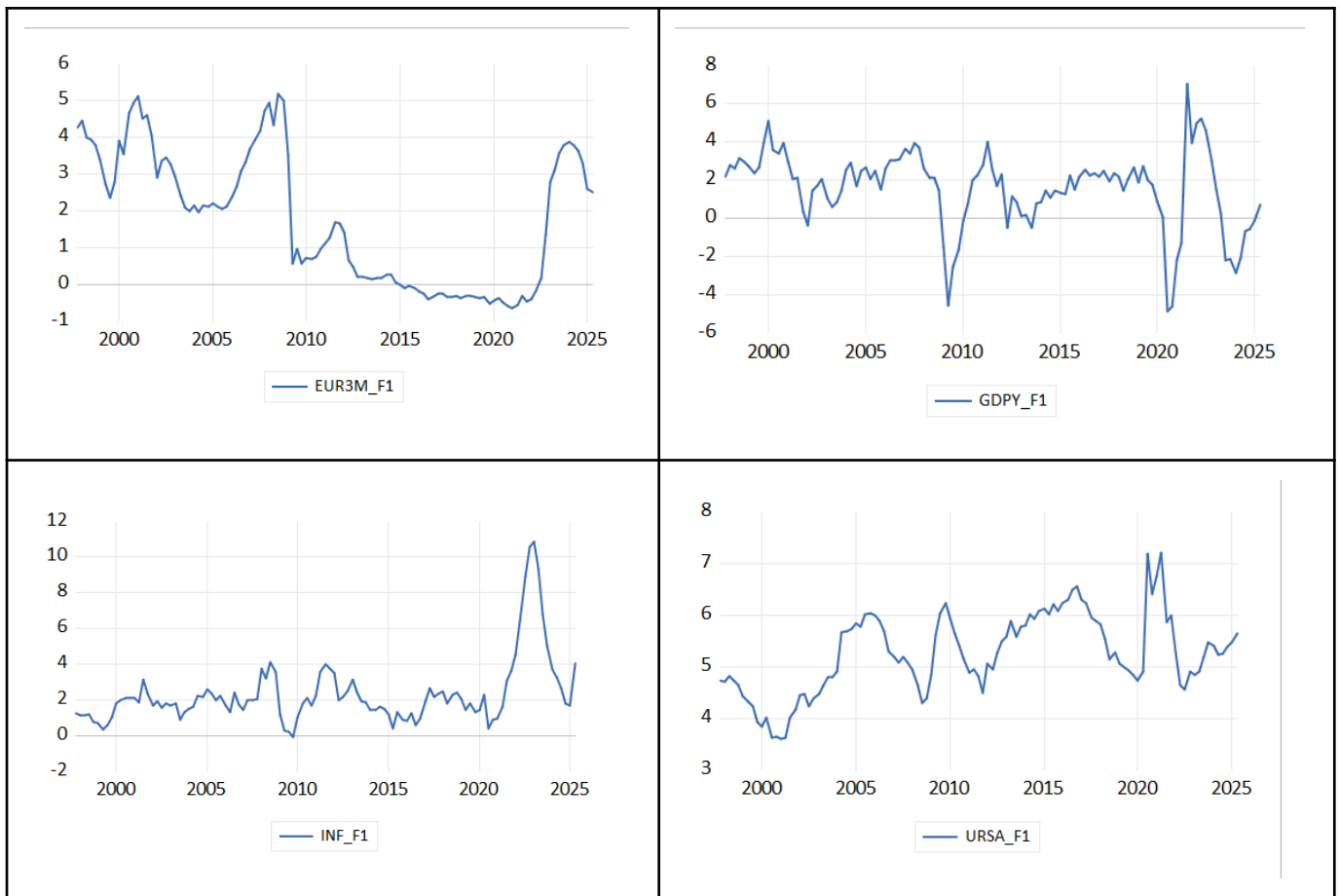
In this part, in endogenous variable we choose: $d(ursa) \text{ inf } gdp \text{ } d(eur3m)$

No change in exogenous variable



A vector autoregression (VAR) model was estimated including the unemployment rate, inflation rate, real GDP growth rate, and the short-term interest rate as endogenous variables. The results indicate meaningful dynamic interdependence among the macroeconomic variables. Lagged unemployment terms have a statistically significant effect on GDP growth, reflecting the cyclical relationship between labor market conditions and output. Inflation shows limited feedback effects on the other variables, while past values of GDP growth exert a strong and statistically significant influence on its own dynamics, highlighting persistence in business cycle fluctuations. Changes in the interest rate respond to past movements in real activity and inflation, suggesting an endogenous monetary policy reaction within the system. The overall fit varies across equations, with relatively higher explanatory power for the GDP growth and interest rate equations. These results confirm the importance of modeling the variables jointly and provide a multivariate framework that captures feedback effects absent in univariate ARIMA models.





The forecasting performance of the VAR model with the interest rate included as an endogenous variable is evaluated using standard accuracy measures, including RMSE, MAE, MAPE, and Theil's inequality coefficient. The results indicate heterogeneous forecasting performance across variables.

For the unemployment rate, the VAR model performs relatively well, with a low RMSE (0.3508) and MAE (0.2393), and a Theil's coefficient of 0.0330, suggesting good predictive accuracy and an improvement over naïve benchmarks. The forecasted unemployment series closely tracks the actual dynamics, capturing medium-term movements reasonably well.

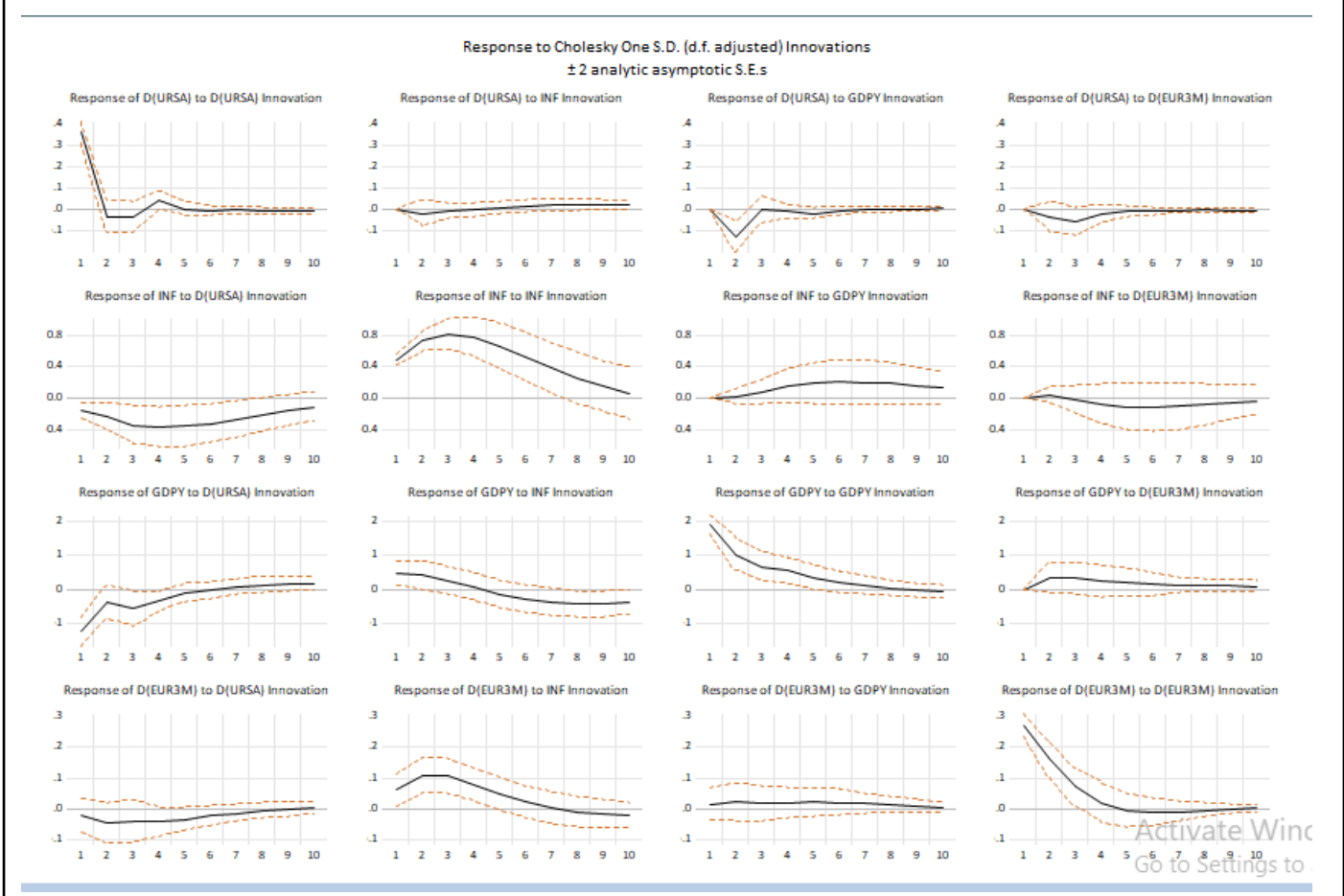
In contrast, the VAR model exhibits weaker forecasting performance for inflation and GDP growth. Inflation forecasts show a higher RMSE (0.4917), Theil's coefficient (0.0819) and a relatively large MAPE, reflecting difficulties in capturing sharp inflationary episodes, particularly during periods of extreme volatility. Similarly, GDP growth forecasts are characterized by a high RMSE (2.2225) and Theil's coefficient (0.3729), indicating limited accuracy in predicting turning points and large economic shocks.

The interest rate forecasts perform moderately, with an RMSE of 0.2687 and a Theil's coefficient of 0.0529. While the model captures the broad movements in interest rates, it struggles during periods of rapid policy shifts, as also evident from the forecast graphs.

Overall, the results suggest that while treating the interest rate as an endogenous variable improves forecasting performance for the unemployment rate, the VAR model remains less effective for forecasting inflation and GDP growth. This highlights the limitations of unrestricted VAR models in the presence of large structural shocks and

motivates the comparison with alternative specifications, including VAR models with exogenous interest rates and Bayesian VAR approaches.

Figure: Impulse Response Dynamics (VAR Model when Interest Rate Endogenous)



The impulse response functions illustrate the dynamic effects of one-standard-deviation shocks to each variable on the rest of the system, based on a Cholesky decomposition with responses shown over a ten-period horizon. Overall, the responses are well behaved and converge to zero, indicating a stable VAR system.

A positive shock to the **unemployment rate** generates a short-lived negative response in **GDP growth**, consistent with standard macroeconomic intuition linking labor market deterioration to lower economic activity. The effect dissipates within a few periods. Inflation and the interest rate exhibit relatively muted responses to unemployment shocks, suggesting limited direct spillovers from labor market disturbances.

An **inflation shock** leads to a gradual decline in **GDP growth** and induces a positive response in the **interest rate**, reflecting a monetary policy tightening reaction to inflationary pressures. The inflation response to its own shock is persistent but mean-reverting, while the effects on unemployment remain small and short-lived.

Shocks to **GDP growth** produce a positive response in the interest rate and a decline in unemployment, highlighting the cyclical interaction between real activity and monetary policy. These effects are strongest in the short run and gradually fade over time.

Finally, an **interest rate shock** results in a temporary reduction in GDP growth and a modest increase in unemployment, consistent with contractionary monetary policy effects. The impact on inflation is limited, suggesting a relatively weak and delayed price channel in the system.

Overall, the impulse response dynamics are economically plausible and reinforce the importance of modeling the macroeconomic variables jointly. While the IRFs provide useful insights into dynamic interactions, they are primarily illustrative in this forecasting-oriented study and complement the forecast evaluation results rather than driving the main conclusions.

Interest Rate Exogenous: The interest rate is treated as exogenous, meaning it affects the other three variables but is not affected by them. This specification assumes interest rate is set by monetary policy based on factors outside our model.

In this part, in endogenous variable we choose: $d(ursa) \text{ inf } gdp_y$

And in exogenous variable we choose: $d(eur3m)$

Var: UNTITLED Workfile: WORKFILE_3RD DEC::q\

View Proc Object Print Name Freeze Estimate Forecast Stats Impulse Resid

Vector Autoregression Estimates

Vector Autoregression Estimates
Date: 12/18/25 Time: 13:09
Sample (adjusted): 1997Q3 2025Q2
Included observations: 112 after adjustments
Standard errors in () & t-statistics in []

	D(URSA)	INF	GDPY
D(URSA(-1))	-0.301701 (0.10886) [-2.77146]	0.039383 (0.15132) [0.26026]	0.943906 (0.68121) [1.38564]
D(URSA(-2))	-0.138544 (0.10959) [-1.26425]	-0.146987 (0.15233) [-0.96493]	-0.532683 (0.68575) [-0.77679]
INF(-1)	0.057647 (0.06611) [0.87202]	1.406567 (0.09189) [15.3068]	0.003322 (0.41368) [0.00803]
INF(-2)	-0.036382 (0.06278) [-0.57954]	-0.535023 (0.08726) [-6.13110]	-0.343167 (0.39284) [-0.87355]
GDPY(-1)	-0.067947 (0.01823) [-3.72748]	0.004576 (0.02534) [0.18060]	0.534133 (0.11407) [4.68258]

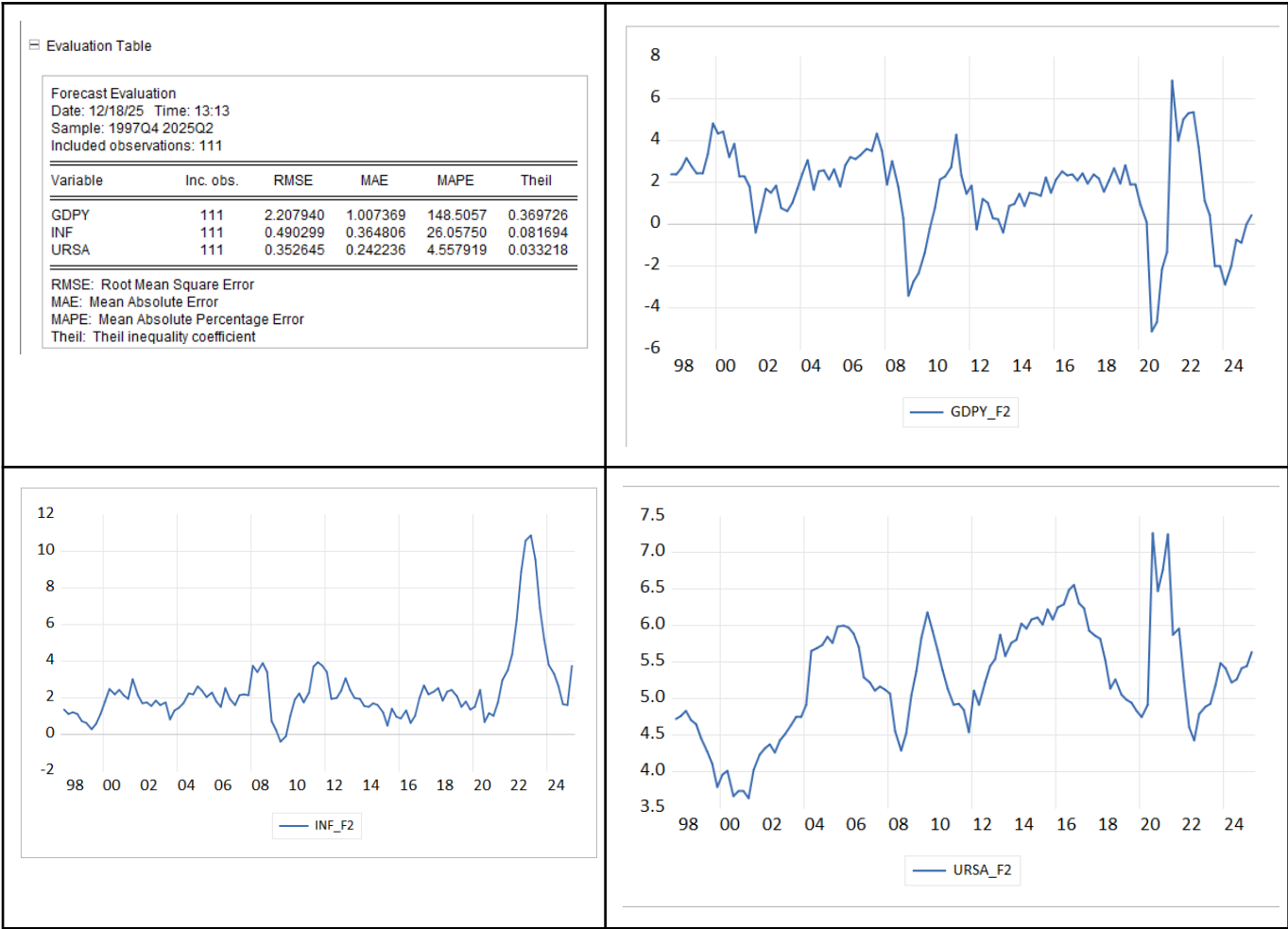
Var: UNTITLED Workfile: WORKFILE_3RD DEC::q\

View Proc Object Print Name Freeze Estimate Forecast Stats Impulse Resid

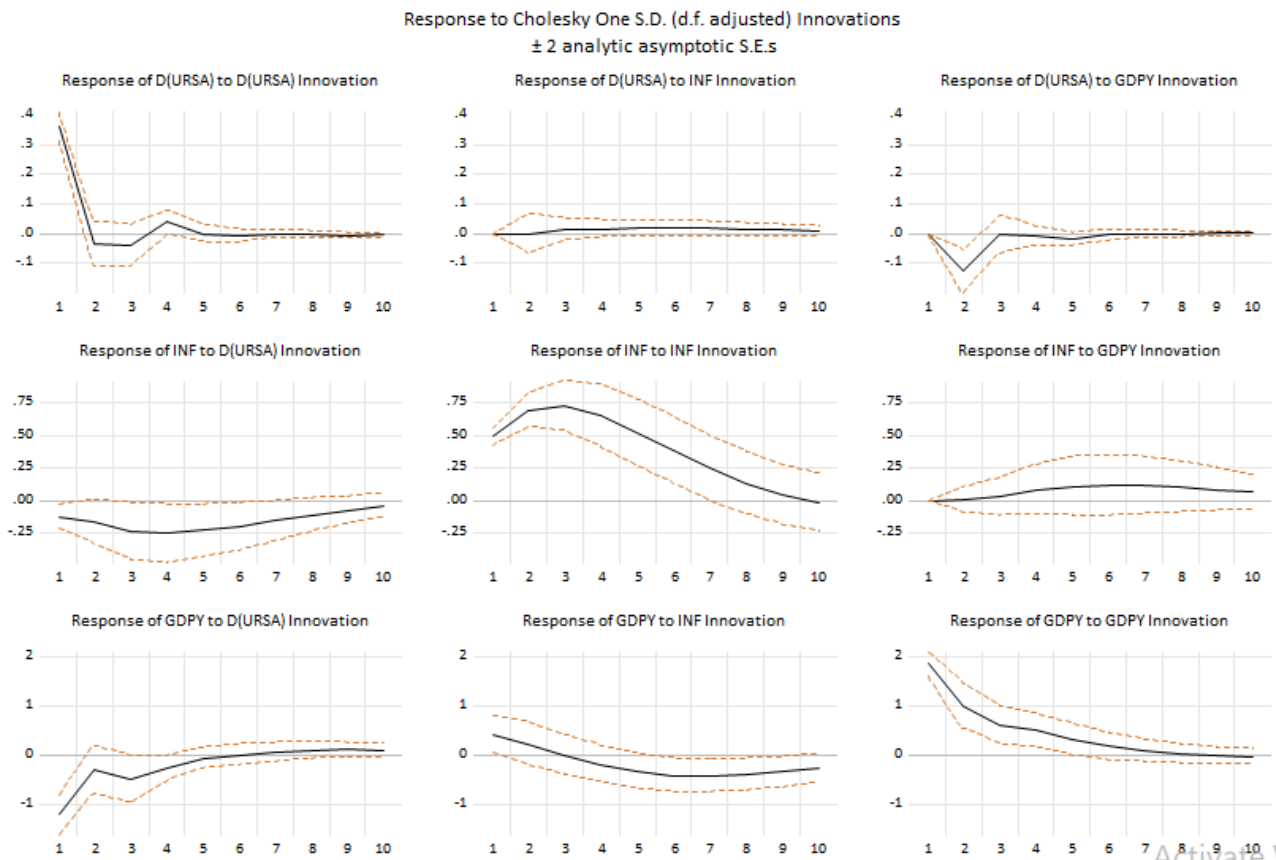
Vector Autoregression Estimates

GDPY(-2)	0.013851 (0.01756) [0.78874]	0.014499 (0.02441) [0.59394]	0.112511 (0.10989) [1.02383]
C	0.045641 (0.06283) [0.72639]	0.279088 (0.08734) [3.19537]	1.351922 (0.39319) [3.43834]
D(EUR3M)	-0.152461 (0.10914) [-1.39695]	0.320142 (0.15171) [2.11027]	1.408287 (0.68295) [2.06207]
R-squared	0.171703	0.935618	0.458284
Adj. R-squared	0.115952	0.931284	0.421823
Sum sq. resids	13.82105	26.70520	541.2074
S.E. equation	0.364547	0.506735	2.281210
F-statistic	3.079835	215.9079	12.56895
Log likelihood	-41.75196	-78.63725	-247.1381
Akaike AIC	0.888428	1.547094	4.556038
Schwarz SC	1.082606	1.741272	4.750216
Mean dependent	0.008631	2.313852	1.591536
S.D. dependent	0.387718	1.933098	3.000093
Determinant resid covariance (dof adj.)	0.113735		
Determinant resid covariance	0.091062		
Log likelihood	-342.5755		
Akaike information criterion	6.545992		
Schwarz criterion	7.128527		
Number of coefficients	24		

A VAR model was estimated for unemployment, inflation, and real GDP growth, treating the short-term interest rate as an exogenous variable. The results indicate that lagged GDP growth plays an important role in explaining the dynamics of unemployment and output, reflecting strong cyclical interactions within the real economy. Inflation exhibits relatively weak spillover effects on the other variables and is mainly driven by its own past values. The exogenous interest rate enters the system with economically meaningful effects, particularly in the GDP growth equation, highlighting the influence of monetary policy on real activity. Compared to the fully endogenous VAR, this specification offers a more parsimonious framework but abstracts from endogenous monetary policy responses.



The forecast evaluation results indicate mixed performance across the macroeconomic variables when the interest rate is treated as an exogenous variable. The model performs relatively well for the unemployment rate, with a low RMSE of 0.3526 and a Theil’s inequality coefficient of 0.0332, indicating good predictive accuracy. Forecasting performance for inflation is moderate, with an RMSE of 0.4903 and a Theil’s coefficient of 0.0817, reflecting difficulties in capturing sharp inflationary movements, particularly in the post-pandemic period. In contrast, GDP growth forecasts exhibit relatively weak accuracy, as indicated by a high RMSE of 2.2079 and a Theil’s coefficient of 0.3697, highlighting challenges in predicting large economic contractions and recoveries. Overall, while treating the interest rate as exogenous yields satisfactory unemployment forecasts, it does not substantially improve forecasting performance for output and inflation compared to alternative model specifications.

Figure: *Impulse Response Dynamics (VAR Model when Interest Rate Exogenous)*

The impulse response functions illustrate the dynamic interactions among unemployment, inflation, and real GDP growth following one-standard-deviation shocks, with the interest rate treated as exogenous. A positive shock to **unemployment** leads to a short-run decline in **GDP growth**, consistent with standard business-cycle dynamics, while its effect on inflation is small and short-lived. An **inflation shock** generates a persistent but gradually declining response in inflation itself and induces a temporary reduction in GDP growth, indicating adverse real effects of inflationary pressures. Shocks to **GDP growth** produce a positive own response that decays over time and are associated with a gradual decline in unemployment, reflecting improved labor market conditions during expansions. Overall, the responses converge smoothly to zero, suggesting a stable VAR system and economically plausible dynamic relationships, though the exclusion of endogenous monetary policy limits feedback effects within the system.

3.3 Bayesian VAR Models

BVAR models are similar to VAR models but use a Bayesian approach. Instead of just using the data, Bayesian methods incorporate prior beliefs about the parameters before seeing the data, then update these beliefs based on the observed data.

The most common prior for BVAR is the Minnesota/Litterman prior, which embodies several sensible beliefs:

- Each variable's recent past values are more important than other variables' past values
- Recent lags matter more than distant lags
- Variables behave somewhat like random walks (their best forecast is their current value)

Advantage: BVAR models often forecast better than unrestricted VARs because the priors prevent overfitting and stabilize estimates, especially valuable with limited data.

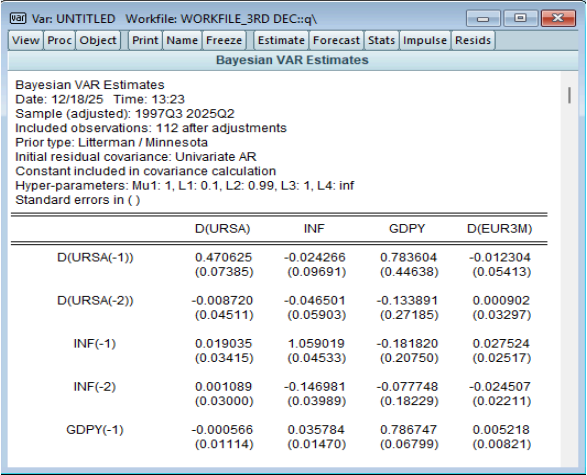
Technical note: The prior is controlled by hyperparameters that determine how tightly the estimates are pulled toward the prior beliefs versus being determined entirely by the data. In our analysis, standard hyperparameter values were used.

Two BVAR Specification:

All Endogenous: All four variables (unemployment, inflation, GDP growth, interest rate) are treated as endogenous, meaning they all influence each other.

In this part, in endogenous variable we choose: *d(ursa) inf gdpd d(eur3m)*

No change in exogenous variable



	D(URSA)	INF	GDPY	D(EUR3M)
D(URSA(-1))	0.470625 (0.07385)	-0.024266 (0.09691)	0.783604 (0.44638)	-0.012304 (0.05413)
D(URSA(-2))	-0.008720 (0.04511)	-0.046501 (0.05903)	-0.133891 (0.27185)	0.000902 (0.03297)
INF(-1)	0.019035 (0.03415)	1.059019 (0.04533)	-0.181820 (0.20750)	0.027524 (0.02517)
INF(-2)	0.001089 (0.03000)	-0.146981 (0.03989)	-0.077748 (0.18229)	-0.024507 (0.02211)
GDPY(-1)	-0.000566 (0.01114)	0.035784 (0.01470)	0.796747 (0.06799)	0.005218 (0.00821)

GDPY(-2)	0.002782 (0.00721)	0.009497 (0.00952)	0.003969 (0.04417)	-0.000206 (0.00532)
D(EUR3M(-1))	-0.084313 (0.09099)	0.103133 (0.12008)	0.539868 (0.55287)	0.779064 (0.06739)
D(EUR3M(-2))	-0.016453 (0.05987)	-0.095263 (0.07900)	-0.017846 (0.36375)	-0.054897 (0.04448)
C	-0.046116 (0.06514)	0.145074 (0.08598)	0.923026 (0.39588)	-0.022810 (0.04804)
R-squared	-0.232238	0.915610	0.407412	0.443871
Adj. R-squared	-0.327946	0.909055	0.361385	0.400676
Sum sq. resid	20.56125	35.00448	592.0324	8.858235
S.E. equation	0.446793	0.582966	2.397475	0.293261
F-statistic	-2.426535	139.6897	8.851717	10.27609
Mean dependent	0.008631	2.313852	1.591536	-0.019812
S.D. dependent	0.387718	1.933098	3.000093	0.378812

The estimated dynamics indicate persistence in GDP growth and inflation, while the interest rate responds systematically to past movements in real activity and prices, reflecting endogenous monetary policy behavior. Compared to the standard VAR, the coefficients appear more stable and economically plausible, with reduced noise across equations. Overall, the BVAR provides a more parsimonious and robust multivariate framework, which is expected to improve out-of-sample forecasting performance.

Var: UNTITLED Workfile: WORKFILE_3RD DEC::q\

ViewProcObjectPrintNameFreezeEstimateForecastStatsImpulseResids

Forecast Evaluation
Date: 12/18/25 Time: 13:28
Sample: 1997Q4 2025Q2
Included observations: 111

Variable	Inc. obs.	RMSE	MAE	MAPE	Theil
EUR3M	111	0.282365	0.160081	14.19290	0.055436
GDPY	111	2.309448	1.040920	139.5398	0.374239
INF	111	0.561395	0.396151	28.62301	0.094011
URSA	111	0.430308	0.292626	5.462742	0.040504

RMSE: Root Mean Square Error
MAE: Mean Absolute Error
MAPE: Mean Absolute Percentage Error
Theil: Theil inequality coefficient

The forecast evaluation results indicate that the Bayesian VAR model with the interest rate treated as an endogenous variable delivers mixed but generally improved forecasting performance relative to the standard VAR. The model performs particularly well for the unemployment rate, with a low RMSE of 0.4303 and a Theil’s inequality coefficient of 0.0405, indicating strong predictive accuracy. Forecasts for the interest rate also improve, as reflected by an RMSE of 0.2824 and a Theil’s coefficient of 0.0554, suggesting better tracking of interest rate dynamics. In contrast, forecasting performance for GDP growth remains relatively weak, with a high RMSE of 2.3094 and a Theil’s coefficient of 0.3742, reflecting continued difficulty in predicting large output shocks. Inflation forecasts show moderate accuracy, with an RMSE of 0.5614 and a Theil’s coefficient of 0.0940, indicating some improvement over the unrestricted VAR but still limited performance during volatile periods.



The impulse response functions from the Bayesian VAR model illustrate smoother and more stable dynamic responses compared to the standard VAR, reflecting the effect of Bayesian shrinkage. A positive shock to unemployment leads to a temporary decline in GDP growth and a gradual adjustment in inflation, with effects dissipating over time. Inflation shocks generate a persistent own response and induce a contractionary response in GDP growth, while also triggering a gradual increase in the interest rate, consistent with an endogenous monetary policy reaction. Shocks to GDP growth result in declining unemployment and a positive response of the interest rate, highlighting cyclical interactions between real activity and monetary policy. Overall, the responses converge smoothly to zero, indicating a stable system and economically plausible dynamics, with less volatility than in the unrestricted VAR, supporting the suitability of the BVAR framework for forecasting purposes.

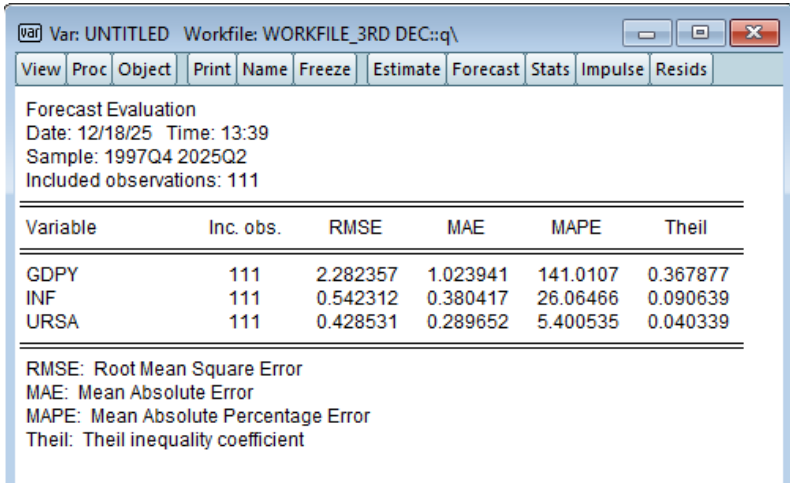
Interest Rate Exogenous: The interest rate is treated as exogenous, meaning it affects the other three variables but is not affected by them. This specification assumes interest rate is set by monetary policy based on factors outside our model.

In this part, in endogenous variable we choose: $d(ursa) \text{ inf } gdp$

And in exogenous variable we choose: $d(eur3m)$

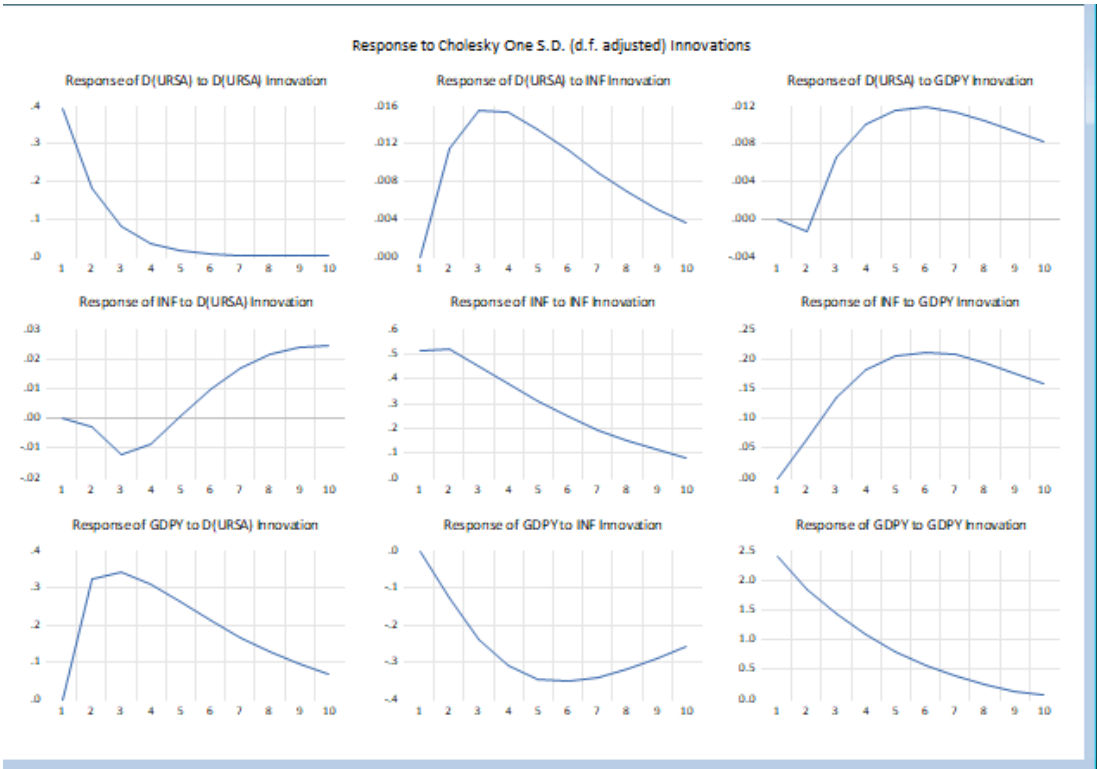
Bayesian VAR Estimates			
Date: 12/18/25 Time: 13:33			
Sample (adjusted): 1997Q3 2025Q2			
Included observations: 112 after adjustments			
Prior type: Litterman / Minnesota			
Initial residual covariance: Univariate AR			
Constant included in covariance calculation			
Hyper-parameters: Mu1: 1, L1: 0.1, L2: 0.99, L3: 1, L4: inf			
Standard errors in ()			
	D(URSA)	INF	GDPY
D(URSA(-1))	0.458179 (0.07325)	-0.006869 (0.09534)	0.816057 (0.44496)
D(URSA(-2))	-0.009045 (0.04494)	-0.042740 (0.05833)	-0.134845 (0.27219)
INF(-1)	0.022463 (0.03474)	1.011569 (0.04574)	-0.246284 (0.21209)
INF(-2)	-0.002934 (0.03019)	-0.137835 (0.03982)	-0.046594 (0.18434)
GDPY(-1)	-0.000517 (0.01108)	0.026734 (0.01449)	0.772796 (0.06795)
GDPY(-2)	0.002739 (0.00715)	0.008163 (0.00936)	0.003474 (0.04402)
C	-0.045590 (0.06322)	0.260593 (0.08275)	1.037657 (0.38608)
D(EUR3M)	-0.126190 (0.11133)	0.568498 (0.14573)	1.184645 (0.67981)
R-squared	-0.222229	0.921206	0.421201
Adj. R-squared	-0.304495	0.915902	0.382243
Sum sq. resids	20.39424	32.68326	578.2562
S.E. equation	0.442830	0.560591	2.357998
F-statistic	-2.701371	173.6990	10.81176
Mean dependent	0.008631	2.313852	1.591536
S.D. dependent	0.387718	1.933098	3.000093

The use of Bayesian shrinkage reduces parameter uncertainty and yields smoother coefficient estimates compared to the standard VAR, improving model stability. The results indicate strong persistence in GDP growth and inflation, while unemployment dynamics are primarily driven by real activity rather than inflationary shocks. The exogenous interest rate enters the system with economically meaningful effects, particularly in the GDP growth equation, highlighting the role of monetary policy in influencing real economic activity. Overall, this specification provides a parsimonious and robust multivariate framework, suitable for forecasting when monetary policy is assumed to be weakly exogenous.



The forecast evaluation results for the BVAR model with the interest rate treated as an exogenous variable indicate heterogeneous performance across variables. The model performs well for the unemployment rate, with a relatively low RMSE of 0.4285 and a Theil’s inequality coefficient of 0.0403, suggesting good predictive accuracy. Forecasting performance for inflation is moderate, with an RMSE of 0.5423 and a Theil’s coefficient of 0.0906, reflecting difficulties in capturing periods of elevated inflation volatility. In contrast, GDP growth forecasts remain relatively weak, as indicated by a high RMSE of 2.2824 and a Theil’s coefficient of 0.3679, highlighting the persistent challenge of predicting large output fluctuations. Overall, treating the interest rate as exogenous in the BVAR framework improves forecast stability but does not substantially enhance accuracy for GDP growth relative to alternative specifications.

Figure: *Impulse Response Dynamics (BVAR Model when Interest Rate Exogenous)*



The impulse response functions from the Bayesian VAR model illustrate the dynamic interactions among unemployment, inflation, and real GDP growth when the interest rate is treated as exogenous. A positive shock to unemployment leads to a temporary decline in GDP growth, while inflation responds only modestly, indicating limited spillover effects from labor market disturbances. Inflation shocks generate a persistent own response and induce a gradual reduction in GDP growth, reflecting adverse real effects of rising prices. Shocks to GDP growth result in a positive own response that gradually decays over time and are associated with declining unemployment, highlighting standard business-cycle dynamics. Overall, the responses are smooth, mean-reverting, and converge to zero, indicating a stable system; however, the exclusion of endogenous monetary policy dampens feedback effects compared to the BVAR specification with an endogenous interest rate.

3.4 Forecast Evaluation Metrics

To compare forecast accuracy across models, we use several standard metrics:

1. Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{[\sum (Actual - Forecast)^2 / n]}$$

RMSE measures the average magnitude of forecast errors, giving more weight to large errors. Lower values indicate better forecasts. For example, an RMSE of 0.5 for unemployment means the typical forecast error is about 0.5 percentage points.

2. Mean Absolute Error (MAE):

$$MAE = \sum |Actual - Forecast| / n$$

MAE measures the average absolute forecast error, treating all errors equally regardless of size. It's more intuitive than RMSE—it directly tells you the average size of your forecast mistakes.

3. Theil's Inequality Coefficient (Theil's U):

We compute Theil's inequality coefficient as $U = RMSE / RMS(actual) = \sqrt{(\sum (F-A)^2 / n)} / \sqrt{(\sum A^2 / n)}$. This measure is scale-free. Values close to zero indicate good forecasting; values near or above 1 indicate poor performance relative to the naïve benchmark of predicting zero.

4. Direction of Change :

To assess whether each model correctly predicts the *sign* of quarterly movements in the variables, I compute a Direction of Change (DoC) statistic. For each forecast period t , I compare the sign of the actual change with the sign of the forecasted change:

$$\Delta y_t^{\text{actual}} = y_t - y_{t-1}, \quad \Delta y_t^{\text{forecast}} = \hat{y}_t - y_{t-1}$$

$$\text{DoC} = \frac{1}{N} \sum_{t=1}^N 1 [\text{sign}(\Delta y_t^{\text{actual}}) = \text{sign}(\Delta y_t^{\text{forecast}})]$$

Observations with zero actual change were excluded so that the statistic captures true directional movements. The DoC statistic ranges from 0 to 1, where values above 0.5 indicate that a model predicts direction better than random guessing. This is particularly important for policy decisions where knowing the direction of change matters more than the exact magnitude.

In Eviews, I ran the following command for Unemployment rate (ursa):

```
series actual_ursa = ursa
series forecast_ursa = ursaf
series actual_ursa_chg = actual_ursa - actual_ursa(-1)
series forecast_ursa_chg = forecast_ursa - forecast_ursa(-1)
series direction_ursa = ((actual_ursa_chg > 0 and forecast_ursa_chg > 0) or (actual_ursa_chg < 0 and forecast_ursa_chg < 0))
scalar DoC_ursa = @mean(direction_ursa)*100
show DoC_ursa
```

I ran the following command for Inflation rate (inf):

```
series actual_inf = inf
series forecast_inf = inff
series actual_inf_chg = actual_inf - actual_inf(-1)
series forecast_inf_chg = forecast_inf - forecast_inf(-1)
series direction_inf = ((actual_inf_chg > 0 and forecast_inf_chg > 0) or (actual_inf_chg < 0 and forecast_inf_chg < 0))
scalar DoC_inf = @mean(direction_inf)*100
show DoC_inf
```

I ran the following command for GDP growth rate (gdpy):

```
series actual_gdpy = gdpy
series forecast_gdpy = gdpyf
series actual_gdpy_chg = actual_gdpy - actual_gdpy(-1)
series forecast_gdpy_chg = forecast_gdpy - forecast_gdpy(-1)
series direction_gdpy = ((actual_gdpy_chg > 0 and forecast_gdpy_chg > 0) or (actual_gdpy_chg < 0 and forecast_gdpy_chg < 0))
scalar DoC_gdpy = @mean(direction_gdpy)*100
show DoC_gdpy
```

I ran the following command for interest rate (eur3m):

```

series actual_eur3m = eur3m
series forecast_eur3m = eur3mf
series actual_eur3m_chg = actual_eur3m - actual_eur3m(-1)
series forecast_eur3m_chg = forecast_eur3m - forecast_eur3m(-1)
series direction_eur3m = ((actual_eur3m_chg > 0 and forecast_eur3m_chg > 0) or (actual_eur3m_chg < 0 and
forecast_eur3m_chg < 0))
scalar DoC_eur3m = @mean(direction_eur3m)*100
show DoC_eur3m

```

I did the same for other variables:

For Unemployment rate (ursa), *forecast_arima (ursaf)*,

forecast_varendogenous(ursa_f1), forecast_varexogenous(ursa_f2),
forecast_bvarendogenous(ursa_f3), forecast_bvarexogenous(ursa_f4).

For Inflation rate (inf), *forecast_arima (inff)*,

forecast_varendogenous(inf_f1), forecast_varexogenous(inf_f2),
forecast_bvarendogenous(inf_f3), forecast_bvarexogenous(inf_f4).

For GDP growth rate (gdpy), *forecast_arima (gdpyf)*,

forecast_varendogenous(gdpy_f1), forecast_varexogenous(gdpy_f2),
forecast_bvarendogenous(gdpy_f3), forecast_bvarexogenous(gdpy_f4).

For interest rate (eur3m), *forecast_arima (eur3mf)*,

forecast_varendogenous(eur3m_f1), forecast_varexogenous(eur3m_f2),
forecast_bvarendogenous(eur3m_f3), forecast_bvarexogenous(eur3m_f4).

Table: Direction of Change Statistics (%)

Percentage of times the model correctly predicted the direction of change

Variable	ARIMA	VAR (endogenous)	VAR (exogenous)	BVAR (endogenous)	BVAR (exogenous)
Unemployment	65.57	61.47	62.29	55.73	55.73
Inflation	56.75	60.52	59.64	59.64	61.40
GDP Growth	64.95	64.10	65.81	65.81	67.52
Interest Rate	76.76	77.46	N/A	78.16	N/A

Note: Direction of Change measures the percentage of times the forecast correctly predicted whether the variable would increase or decrease compared to the previous period. Values above 50% indicate better-than-random performance. N/A indicates the variable was treated as exogenous and therefore not forecast.

Multivariate and Bayesian models tend to outperform ARIMA in predicting the direction of change for variables with stronger cross-variable interactions, particularly GDP growth and interest rates. The largest gains are observed for real GDP growth and the interest rate, where VAR and especially BVAR models achieve higher directional accuracy, reflecting the importance of cross-variable information and Bayesian shrinkage. For unemployment and inflation, improvements are more modest, consistent with their relatively persistent dynamics and sensitivity to idiosyncratic shocks. The highest directional accuracy is achieved by the BVAR models, particularly for GDP growth and the interest rate. Overall, these results complement the RMSE and Theil's coefficient findings and confirm the advantages of Bayesian multivariate frameworks for forecasting turning points.

4. Forecasting Performance

Table: Unemployment Forecast Accuracy

Model	RMSE	MAE	Theil Inequality	Direction of Change (%)
ARIMA	0.3378	0.2411	0.0318	65.57
VAR (Endogenous)	0.3508	0.2393	0.0330	61.47
VAR (Exogenous)	0.3526	0.2422	0.0332	62.29
BVAR (Endogenous)	0.4303	0.2926	0.0405	55.73
BVAR (Exogenous)	0.4285	0.2897	0.0403	55.73

Table: Inflation Forecast Accuracy

Model	RMSE	MAE	Theil Inequality	Direction of Change (%)
ARIMA	0.4481	0.3234	0.0741	56.75
VAR (Endogenous)	0.4917	0.3673	0.0819	60.52

VAR (Exogenous)	0.4903	0.3648	0.0817	59.64
BVAR (Endogenous)	0.5614	0.3962	0.0940	59.64
BVAR (Exogenous)	0.5423	0.3804	0.0906	61.40

Table: GDP Growth Forecast Accuracy

Model	RMSE	MAE	Theil Inequality	Direction of Change (%)
ARIMA	1.6164	0.8787	0.2603	64.95
VAR (Endogenous)	2.2225	1.0165	0.3729	64.10
VAR (Exogenous)	2.2079	1.0074	0.3697	65.81
BVAR (Endogenous)	2.3094	1.0410	0.3742	65.81
BVAR (Exogenous)	2.2824	1.0239	0.3679	67.52

Table: Interest Rate Forecast Accuracy

Model	RMSE	MAE	Theil Inequality	Direction of Change (%)
ARIMA	0.2739	0.1658	0.0541	76.76
VAR (Endogenous)	0.2687	0.1657	0.0529	77.46
VAR (Exogenous)	-	-	-	-
BVAR (Endogenous)	0.2824	0.1601	0.0554	78.16
BVAR (Exogenous)	-	-	-	-

Summary: Best Models

The forecast evaluation results indicate that ARIMA models achieved the lowest RMSE values for the unemployment rate (0.337) and inflation rate (0.448), highlighting the strong performance of univariate models for relatively persistent macroeconomic variables. For real GDP growth, although ARIMA produced competitive error measures (RMSE of 1.616), multivariate BVAR models achieved higher directional accuracy, suggesting an advantage in capturing turning points. In contrast, interest rate forecasting benefited from multivariate modeling, with the VAR model treating the interest rate as endogenous yielding the lowest RMSE (0.273). Overall, while ARIMA models perform well for smoother series, multivariate and Bayesian models add value for variables with stronger cross-variable interactions, particularly during periods of economic turbulence. The mixed results likely reflect the presence of structural breaks and unprecedented shocks during the 2020–2025 evaluation period, which may have weakened stable cross-variable relationships.

5. Discussion

The forecast evaluation indicates that univariate ARIMA models perform particularly well during the 1997Q4–2025Q2 evaluation period, especially for unemployment and inflation. This strong performance can be attributed to the presence of major structural disruptions, including the COVID-19 pandemic, the energy crisis, and the subsequent inflation surge, which weakened historically stable cross-variable relationships. By relying solely on each variable's own dynamics, ARIMA models were less exposed to such structural instability than multivariate VAR and BVAR frameworks.

In contrast, VAR and BVAR models generally exhibit higher forecast errors for unemployment, inflation, and GDP growth, despite their theoretical advantage of capturing economic interactions. The relatively weaker performance of these models reflects parameter proliferation: even with Bayesian shrinkage, multivariate models estimate substantially more parameters than univariate ARIMA specifications, increasing the risk of overfitting during periods when historical relationships provide limited guidance for future outcomes. As a result, cross-variable dynamics captured in-sample often failed to hold in the forecast period.

However, forecast accuracy alone does not fully capture the value of multivariate models. VAR and BVAR frameworks provide important economic insights through impulse response analysis, revealing transmission mechanisms that ARIMA models cannot identify. In particular, the BVAR impulse responses confirm standard monetary transmission channels, showing that interest rate increases reduce GDP growth and inflation with a delay. These structural insights are valuable for policy analysis and scenario evaluation, even when point forecasts are less accurate.

The treatment of the interest rate as endogenous or exogenous yields mixed results across variables. For inflation and unemployment, endogenous specifications generally perform slightly better, suggesting that feedback effects and expectations channels matter even for a small Eurozone economy. In contrast, GDP growth forecasts are largely insensitive to the interest rate specification, indicating that output dynamics are dominated by broader real shocks rather than monetary policy details.

An important exception to ARIMA's dominance is interest rate forecasting, where the VAR model with endogenous interest rates outperforms ARIMA. This reflects the policy-driven nature of interest rates, which respond systematically to developments in inflation, output, and labor market conditions. Incorporating this information improves interest rate forecasts relative to purely autoregressive approaches.

Finally, all models struggle to forecast GDP growth, with high RMSE and Theil's inequality coefficients across specifications. This underscores the inherent difficulty of predicting output during a period marked by unprecedented economic shocks and rapid structural change. The results highlight the limits of statistical forecasting models when faced with extreme and unanticipated events.

6. Conclusion

This study evaluated the forecasting performance of univariate ARIMA models and multivariate VAR and BVAR models for key Austrian macroeconomic variables—unemployment, inflation, real GDP growth, and the short-term interest rate—over the period 1997Q4–2025Q2. By comparing alternative model specifications and forecast evaluation metrics, including RMSE, MAE, Theil's inequality coefficient, and direction-of-change accuracy, the analysis provides a comprehensive assessment of the relative strengths and limitations of different time-series forecasting approaches.

The results show that **no single model dominates across all variables**. Univariate ARIMA models perform particularly well for unemployment and inflation, delivering the lowest forecast errors and demonstrating robustness during a period characterized by severe structural disruptions. Their strong performance highlights the value of parsimonious models when historical cross-variable relationships are unstable. In contrast, multivariate VAR and BVAR models provide clear advantages for forecasting the interest rate, where incorporating information from other macroeconomic variables improves predictive accuracy by capturing policy reaction dynamics.

For real GDP growth, all models exhibit limited forecasting accuracy, reflecting the inherent difficulty of predicting output during periods of unprecedented economic shocks. While Bayesian VAR models marginally improve directional accuracy for GDP growth, large forecast errors persist across specifications, underscoring the fundamental uncertainty surrounding macroeconomic fluctuations in turbulent environments.

Although multivariate and Bayesian models do not consistently outperform ARIMA in terms of point forecast accuracy, they offer **important complementary insights**. In particular, VAR and BVAR impulse response analyses reveal economically meaningful transmission mechanisms—such as the effects of monetary policy shocks on output and inflation—that cannot be identified within a univariate framework. These features make multivariate models especially valuable for policy analysis and scenario evaluation, even when forecasting performance is weaker.

Overall, the findings suggest that **model choice should be guided by the variable of interest and the purpose of the analysis**. While ARIMA models provide reliable benchmarks for forecasting persistent macroeconomic series, VAR and BVAR models remain indispensable tools for understanding economic interactions and policy transmission. The results emphasize the importance of combining forecast accuracy with economic interpretability in applied macroeconomic forecasting.

7. References

Data: Eurostat

Software: Eviews 12 Student Lite & EViews 14

LLM: ChatGPT 5.2